

Course : Cloud Computing Course**Duration : 2 hours/Day**

Sr. No.	Fundamentals of OS and Networking
1	Process and Thread Management
2	CPU Scheduling (basic)
3	Memory Management & Virtual Memory
4	File System & Permissions
5	I/O Management
6	Virtualization & Hypervisors
7	Resource Allocation in OS
8	Basics of Networking
9	Types of Networks (LAN, WAN, VPN, etc.)
10	Network Devices (Router, Switch, Firewall)
11	IP Addressing (IPv4, IPv6, Subnetting)
12	Protocols (HTTP, HTTPS, TCP/IP, DNS, DHCP, SSH, FTP)
13	Ports and Firewalls
14	DNS Working
15	Cloud Networking (VPC, Subnet, Gateway, Route Table)
16	Load Balancing & CDN
17	Network Security (SSL/TLS, VPN)
Sr. No.	Linux Commands and Working with CMD & PowerShell
18	File & Directory Management
19	File Viewing & Editing
20	User & Permission Management
21	Process Management
22	System Information Commands
23	Network Commands
24	Package Management
25	Logs & Monitoring
26	Archive & Compression
27	Shell Scripting Basics
28	Navigation Commands
29	File Operations
30	System Information Commands
31	Network Commands
32	Disk Management
33	Introduction & Cmdlets
34	File Management Cmdlets
35	Process Management Cmdlets
36	Network Cmdlets
Sr. No.	AWS (Amazon Web Services)
38	Introduction to Cloud & AWS
39	AWS Global Infrastructure
40	Region, Availability Zone, Edge Location
41	AWS Management Console & CLI
42	Compute Services
43	EC2 (Elastic Compute Cloud)
44	AMI, Instance Types, Key Pair, Security Groups
45	Auto Scaling, Load Balancer
46	Storage Services
47	S3 (Simple Storage Service)
48	EBS (Elastic Block Store)
49	EFS (Elastic File System)
50	Glacier (Archive Storage)
51	Database Services
52	RDS (Relational Database Service)

53	DynamoDB, Aurora
54	Networking & Content Delivery
55	VPC (Virtual Private Cloud)
56	Subnet, Route Table, Internet Gateway, NAT
57	Elastic IP, CloudFront, Route 53
58	Identity & Access Management (IAM)
59	Users, Groups, Roles, Policies
60	Monitoring & Logging
61	CloudWatch, CloudTrail
62	Serverless Services
63	Lambda, API Gateway
64	DevOps & Deployment
65	Billing, Cost Management, Free Tier
Sr.No.	Introduction to Containerization (Docker)
67	Docker Architecture
68	Client, Daemon, Registry, Image, Container
69	Installing Docker
70	Basic Docker Commands
71	run, ps, images, pull, push, stop, rm
72	Docker Images
73	Creating and Managing Images
74	Dockerfile (Writing and Building)
75	Docker Containers
76	Run, Start, Stop, Remove Containers
77	Docker Volumes
78	Persistent Storage
79	Docker Networking
80	Bridge, Host, Overlay networks
81	Docker Compose
82	docker-compose.yml file
83	Multi-container Applications
84	Docker Hub & Private Repositories
85	Docker Swarm (Basic Orchestration)
Sr.No.	Kubernetes (K8s)
87	Introduction to Container Orchestration
88	Kubernetes Architecture
89	Master Node, Worker Node, etcd, API Server
90	Key Components
91	Pods, ReplicaSets, Deployments, Services
92	Namespaces & Labels
93	YAML Configuration Files
94	Scaling & Rolling Updates
95	Networking in Kubernetes
96	Storage in Kubernetes
97	Persistent Volumes (PV), Persistent Volume Claims (PVC)
98	Config Maps & Secrets
99	Kubernetes on Cloud (EKS, GKE, AKS)
Sr.No.	Introduction to Jenkins
101	Installing Jenkins
102	Jenkins Architecture & Components
103	Master, Agent, Executor, Node
104	Jenkins Dashboard & Jobs
105	Jenkins Pipelines
106	Freestyle Jobs
107	Declarative vs Scripted Pipeline
108	Jenkins file Basics
109	Stages, Steps, Agents
110	Integration with GitHub/GitLab

111	Build Triggers (Commit-based, Scheduled)
112	Jenkins Plugins
113	Build, Test, Deploy Automation
114	Jenkins with Docker
115	Jenkins with Kubernetes
116	Jenkins Security & User Management
Sr.No.	Introduction to CI/CD Pipeline
118	Continuous Integration vs Continuous Delivery vs Continuous Deployment
119	CI/CD Workflow Stages
120	Code → Build → Test → Deploy → Monitor
121	Version Control Integration (Git)
122	Build Automation Tools
123	Maven, Gradle, npm, etc.
124	Testing Automation
125	Unit Tests, Integration Tests
126	Deployment Strategies
127	Blue-Green, Rolling, Canary Deployments
128	Pipeline as Code
129	Using Jenkinsfile or YAML
130	Containerization in CI/CD
131	Integrating Docker builds
132	Continuous Monitoring Tools
133	Prometheus, Grafana
134	CI/CD in Cloud
135	AWS CodePipeline, GitHub Actions, GitLab CI/CD
Sr.No.	Git (Version Control System) GitHub
137	Introduction to Version Control
138	Centralized vs Distributed VCS
139	Installing and Configuring Git
140	Git Basic Concepts (Repository, Working Directory, Staging Area)
141	Initializing and Cloning Repositories
142	Basic Commands (status, add, commit, log, diff)
143	Branching and Merging
144	Resolving Merge Conflicts
145	Working with Remote Repositories (push, pull, fetch)
146	Undoing Changes (restore, reset, revert)
147	Git Tags and Releases
148	Git Stash, .gitignore File
149	Git Aliases, Git Rebase (Advanced)
150	Git Workflows (Feature Branch, Gitflow, Forking)
151	Introduction to GitHub
152	Creating and Managing Repositories
153	Public vs Private Repositories
154	Forking and Cloning Projects
155	Pull Requests and Code Reviews
156	Merging and Branch Protection Rules
157	Issues and Labels
158	Milestones and Assignees
159	GitHub Actions (CI/CD Workflows)
160	GitHub Pages (Website Hosting)
161	SSH Keys and Authentication
162	Personal Access Tokens
163	Managing Collaborators and Teams